

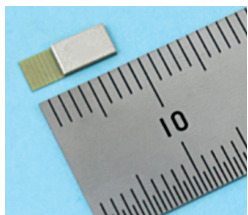
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

World's smallest Bluetooth low energy module

KAGA FEI Co Ltd has launched the ES4L15BA1, claimed to be the world's smallest Bluetooth low energy module. The module is designed for IoT devices,



wearables, and health-care products, where space efficiency and performance are essential.

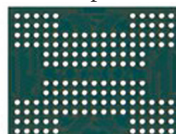
With a built-in antenna and certifications, developers can speed up projects, reducing time-to-market and certification costs. It supports Bluetooth 6.0, including features like channel sounding for distance measurement, making it suitable for applications requiring proximity detection, such as secure device unlocking. Developers can create IoT devices that meet security standards with PSA certification support. Wearable technology companies will find this module valuable for manufacturing compact, Bluetooth-enabled devices.

KAGA FEI
<https://eu.kagafei.com>

World's first 321-layer 4D TLC NAND

SK Hynix Inc. has announced the mass production of the world's first triple-level cell (TLC) 321-layer 4D NAND Flash with 1Tb capacity. The stacking of over 300 layers was achieved using the '3 plugs' process, which connects three plugs electrically through a follow-up process. The company developed a low-stress material to minimise wafer warpage and introduced alignment correction technology for plug processes. Using the development platform of its 238-layer

NAND, the company increased productivity by 59% during the transition. The product provides a 12% improvement



in data transfer speed, a 13% increase in reading performance, and more than 10% improvement in data reading power efficiency compared to the previous generation.

SK Hynix
<https://www.skhynix.com>

World's fastest 60TB data centre SSD

Micron Technology Inc. has launched the 6550 ION NVMe SSD, which the company states are the world's fastest 60TB data centre SSD and the industry's first E3.S and PCIe Gen5 60TB SSD. Designed for exascale data centre deployments, it delivers performance, energy efficiency, endurance, security, and rack density. It is designed for



high-capacity NVMe workloads, such as networked AI data lakes, data preparation, checkpointing,

public cloud storage, analytic databases, and content delivery. The drive offers a 60TB capacity in an E3.S form factor and up to 20% greater energy efficiency than competing solutions. It delivers 179% faster sequential reads, 179% higher read bandwidth per watt, 150% faster sequential writes, 213% higher write bandwidth per watt, 80% faster random reads, and 99% higher read IOPS per watt compared to competitors.

Micron Technology
<https://in.micron.com>

8.5kW power supply powered by GaN and SiC

Navitas Semiconductor has introduced an 8.5kW power supply unit (PSU) powered by GaN and SiC technologies. Designed for AI and hyperscale data centres, the PSU achieves 98% efficiency while meeting Open Compute Project (OCP) and Open Rack v3 (ORv3)



specifications. The design reduces ripple current, EMI, and the number of GaN and SiC devices by 25% compared to competitors, lowering overall cost. The 3-phase LLC topology uses GaN-Safe technology, integrating control, drive, sensing, and protection features, including short-circuit protection, ESD resistance, and programmable slew rate control. The PFC stage is powered by Gen-3 Fast SiC MOSFETs with trench-assisted planar technology.

Navitas Semiconductor
<https://navitassemi.com>

Full-range, near-field silicon microspeaker

xMEMS Labs has introduced xMEMS Sycamore, a full-range, all-silicon, near-field microspeaker. The company claims that the speaker is designed for manufacturers and developers of compact, mobile, and wearable electronics, including consumer electronics companies creating open wireless stereo (OWS) earbuds, smartwatches, and smart glasses; AR/VR device developers building augmented and virtual reality headsets; and laptop and portable